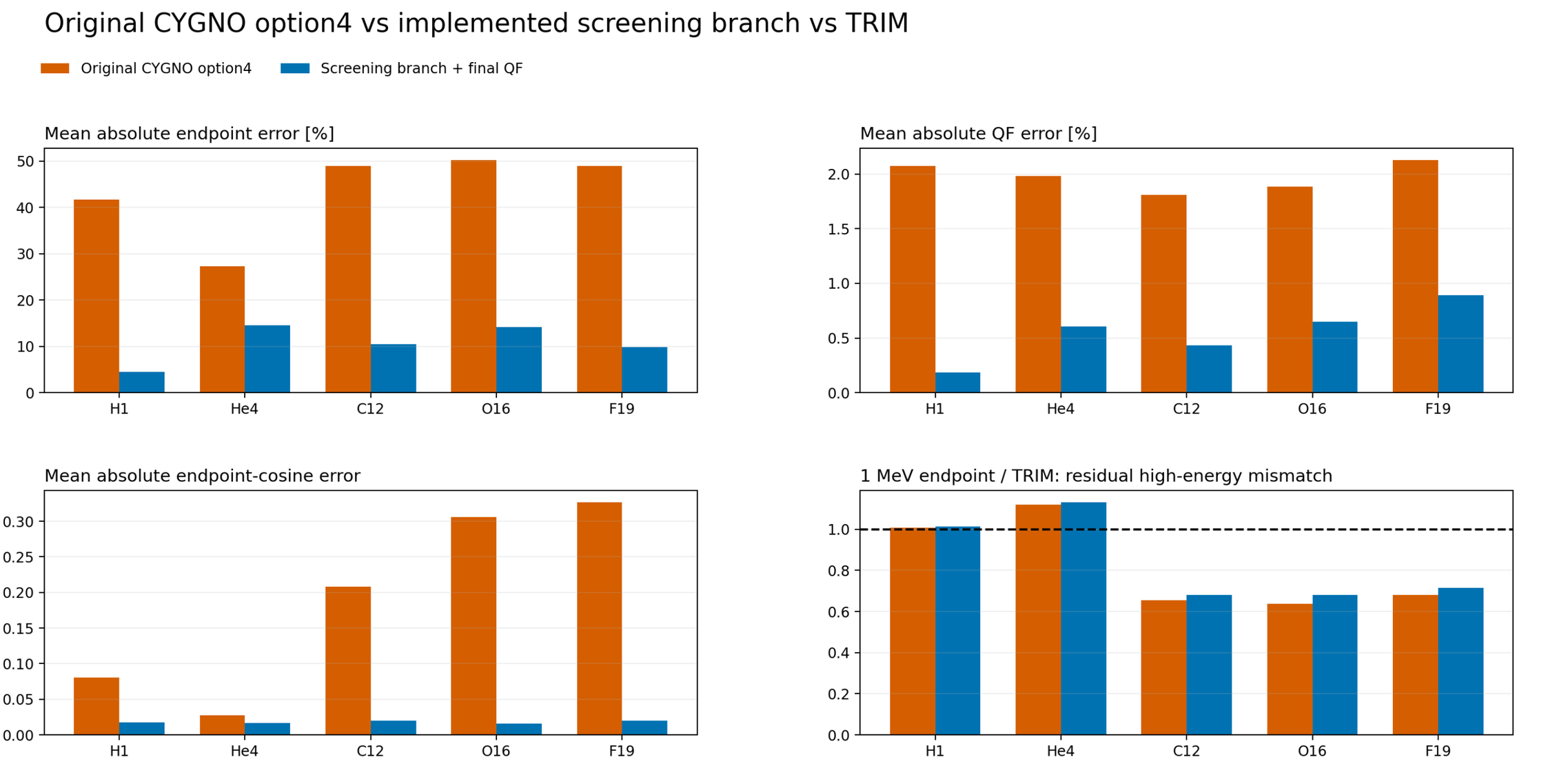
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Comparison overview



The implemented screening reduces the equal-weight mean endpoint discrepancy from 43.38% to 10.69% across five ions and five energies. QF includes the final dense bounded event calibration, not a screening-only prediction. C/O/F endpoints remain 28-32% short at 1 MeV. Scores are descriptive discrepancies, not significance-weighted goodness-of-fit statistics. Standard means the original CYGNO option4 configuration, not every Geant4 physics list.

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Models, samples and uncertainty

Branch comparison: g4v11.3_cygno04 at 3f53d24 versus g4v11.3_cygno04_screening implementation commit 203c22e. The naming convention says 11.3, but all reported builds and transport tests used Geant4 11.4.0 and ROOT 6.36.08 under WSL.

Matched physics sample: 200 events at each of 0.25, 1, 10, 100 and 1000 keV, for each of H1, He4, C12, O16 and F19: 5,000 scored events/model. Additional 25 and 100 eV diagnostics are excluded from all scores. Initial energies and directions match exactly between the original and screened controls. Isotropic single ions start at (0,0,245) mm in the existing 900 mbar, 293.15 K He/CF4 gas.

The archived original control source matches the base branch after LF/CRLF normalization. A 7,000-event integration replay retained the actual new branch physics and event action: every primary tracking field matched the selected screened prototype; raw/native response matched bitwise; the embedded QF matched the frozen calibration to numerical tolerance. The 100-event full-production ROOT smoke test verified scalar/vector response sums. Replays are integration checks, not extra independent physics statistics.

Physics: option4 electronic ionisation is retained. Continuous nuclearStopping and low-energy ion multiple/Coulomb scattering are replaced by explicit ZBL screened collisions for GenericIon, alpha, He3, proton, deuteron and triton. Transfer cutoff: 0.1 eV. Recoil cutoff: 1 eV/u. Existing 25 eV tracking threshold and energy-dependent step caps are unchanged. Two local TestEm7 numerical fixes continue the ZBL potential beyond the interpolation domain and make the terminal-energy tests consistent.

Custom response: the existing local TRIM derivative remains, followed by f_new=1-(1-f_old)^alpha_species(E0). The gain uniformly scales custom event/hit response weights. Raw energies and normalized charge geometry are unchanged; response fluctuations change. Calibration applies only to eligible single-primary H/He/C/O/F events in 0.25-1000 keV and was qualified for contained events in this gas configuration.

The final alpha table was trained on 42,000 fresh events: 28 nodes x 300 events x 5 ions. Fixed-energy comparison points include fit-node energies, although their events are independent of training. Fresh validation adds 20,250 midpoint events and 5,000 log-uniform spectrum events. The rejected coarse correction and previous screened candidate are excluded from the comparison curves.

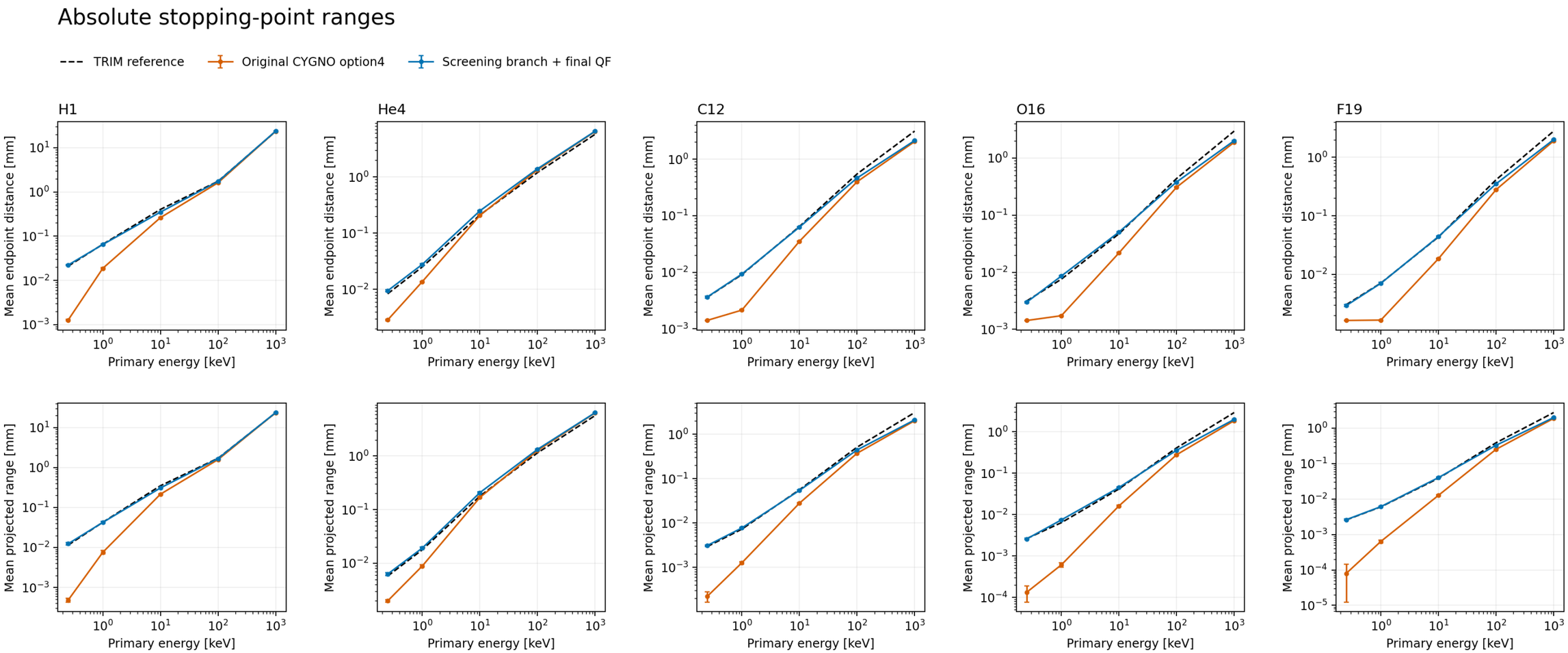
Frozen TRIM range reference: 195 energy/species points from 897 batches and 89,700 ions. Endpoint distance, longitudinal projection, endpoint cosine, backward fraction and stopping-point spread are distinct observables. TRIM electronic energy includes excitation; it is not measured collected charge.

Errors: Geant4 means use event SEM; range ratios combine Geant4 and TRIM mean SEM. QF bars use MC plus conditional fitted-alpha uncertainty where stated. Backward fractions use Wilson 95% intervals. Spread plots have no inferred uncertainty bands. Interpolation/model systematic errors are excluded. No event-level TRIM yield-fluctuation reference is available.

This report supersedes the earlier same-named development PDF for this branch. That report described a previous candidate, a different QF calibration, 100 events/point and prototype packed output. Here the data and labels correspond to the final implemented transport/calibration. The accompanying technical note documents exact code changes.

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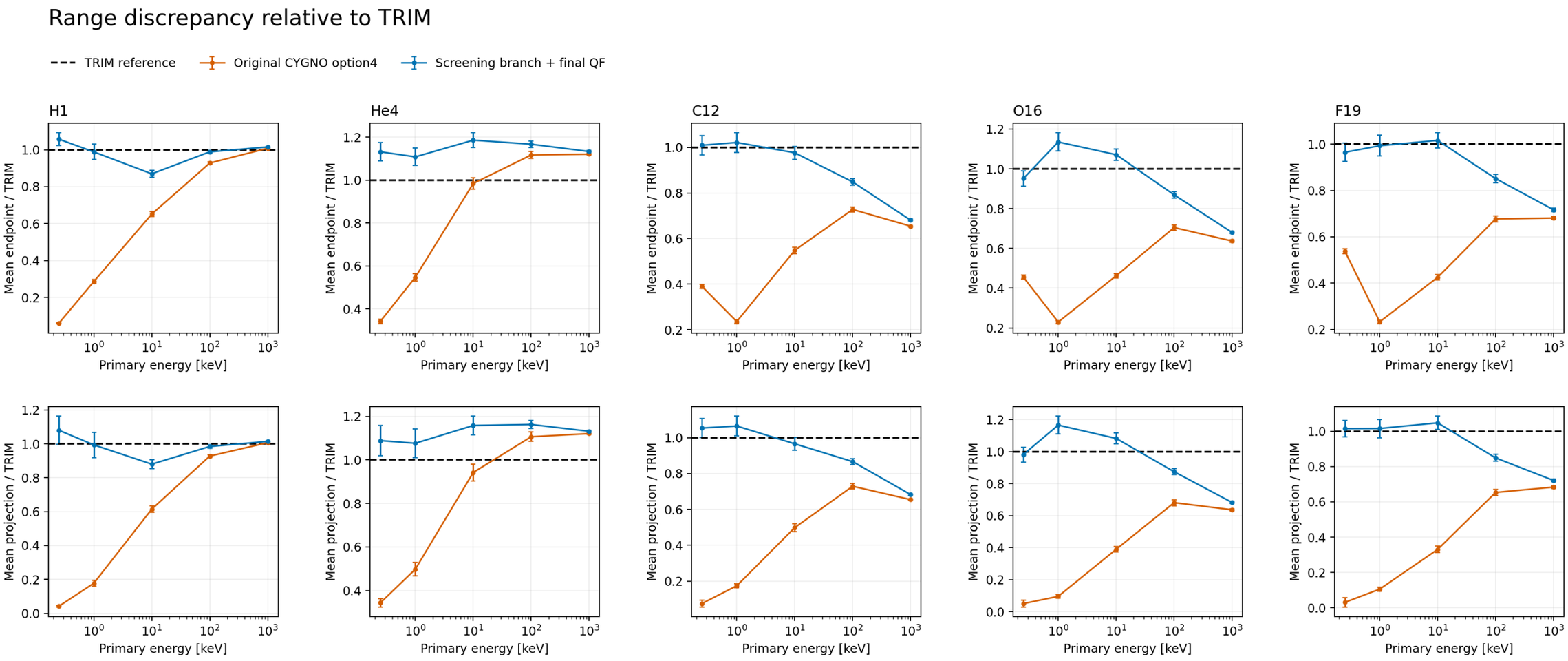
Ranges: absolute endpoint distance and projection



All scored primary tracks stop in sensitive gas. Endpoint distance is the magnitude of displacement from source; projected range is displacement along the initial direction. Error bars show Geant4 event SEM. TRIM reference means are shown as dashed curves. True Geant4 curved path length is not equated to sampled TRIM polyline length.

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Ranges: ratios to TRIM

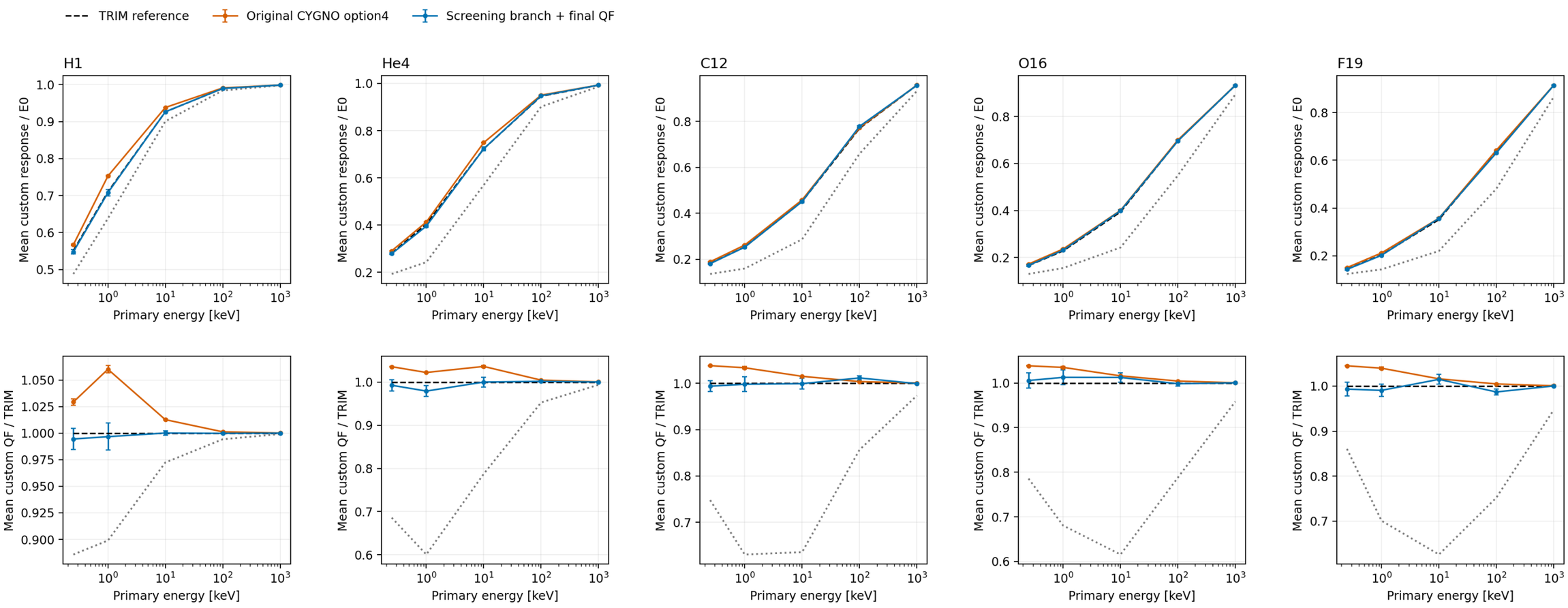


Unity denotes agreement with the TRIM mean. Error bars combine measured Geant4 and TRIM finite-sample uncertainties; reference interpolation and model systematics are excluded. Screening improves overall agreement, but not every energy point. The high-energy heavy-ion discrepancy remains and cannot be repaired by event QF calibration.

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Quenching and the empirical TRIM correction

Quenching: recorded custom response and TRIM

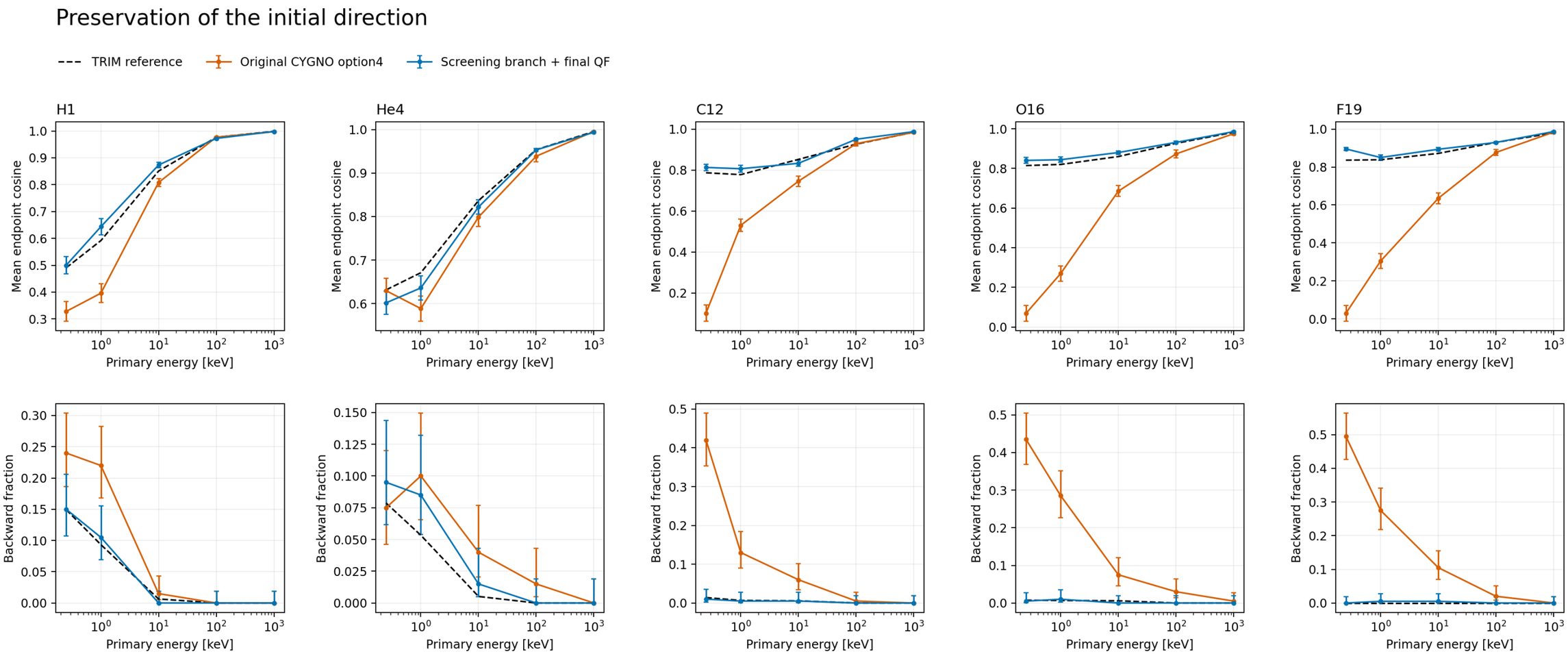


Dotted gray: identical screened transport before the empirical event correction. Blue: implemented branch response.

The blue screening result includes the final 28-node bounded mean-response correction. Gray shows the uncorrected screened response: the mean agreement is not intrinsic to screening alone. The original branch already uses its local TRIM derivative. Error bars combine event SEM and conditional calibration SEM; interpolation/model errors are excluded. No measured collected-charge calibration is claimed.

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Bending and backward stopping points

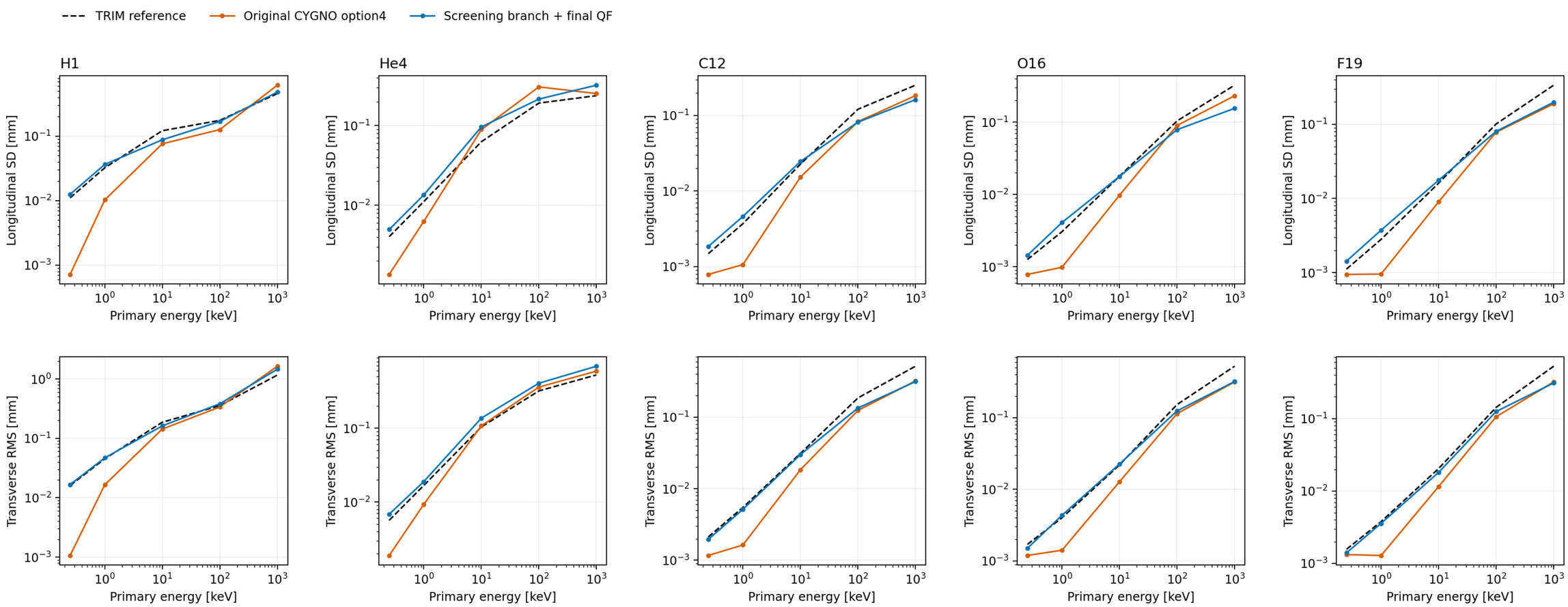


Endpoint cosine uses the initial momentum direction and source-to-stop displacement, not final momentum or reconstructed charge axis. Backward means negative projected displacement. Cosine bars are Geant4 event SEM; backward bars are Wilson 95% intervals. TRIM curves show reference means. Zero backward observations do not establish zero probability.

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Longitudinal and transverse straggling

Stopping-point straggling

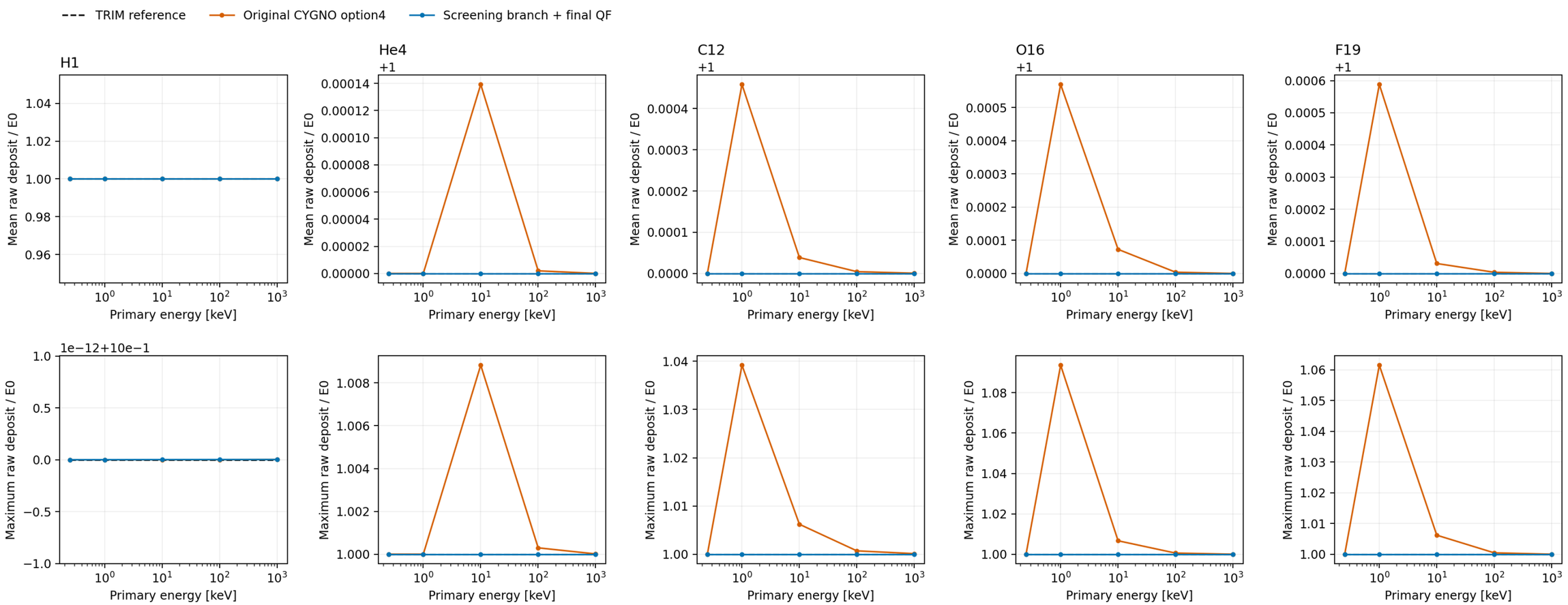


TRIM spread references are reconstructed from endpoint moments, cohort sizes and mean SEMs. Longitudinal SD is the standard deviation of projected endpoint displacement. Transverse RMS is the root-mean-square transverse endpoint distance. These descriptive spread curves have no uncertainty bands on this page. A better mean range does not independently validate the full recoil-cascade morphology.

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Deposited energy and event budgets

Raw event-energy accounting

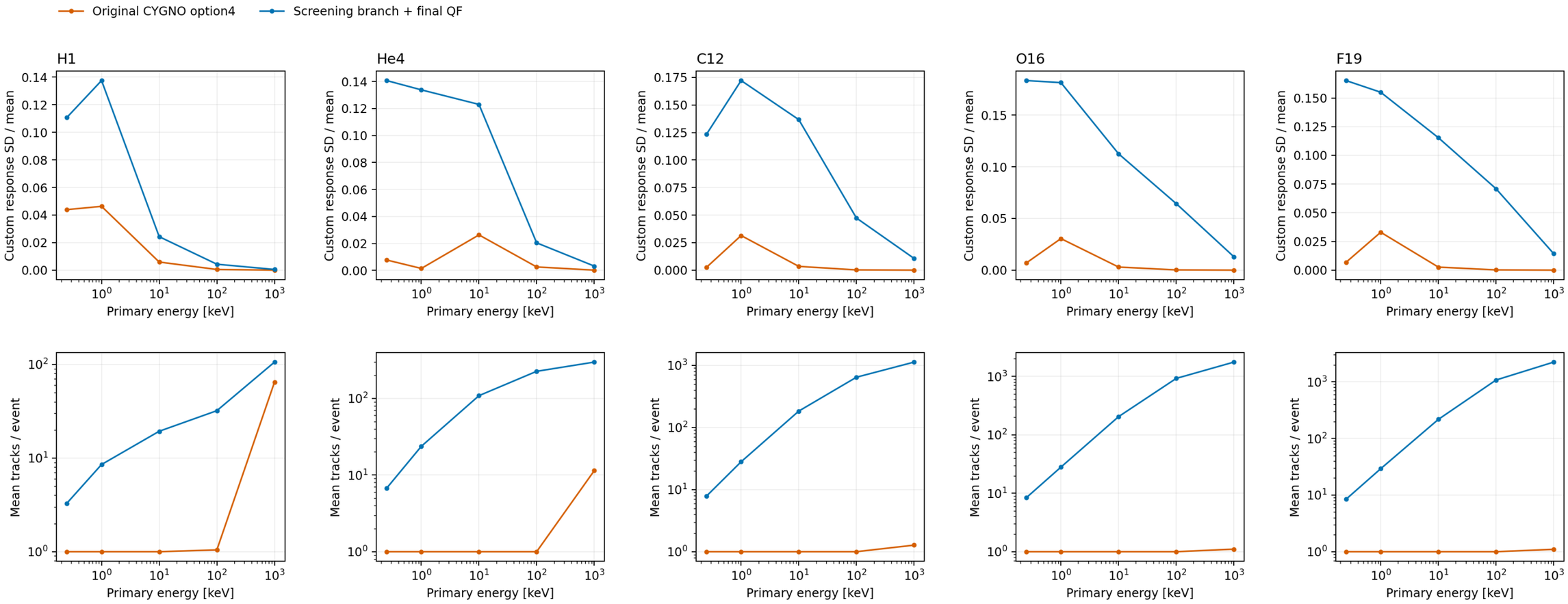


Raw means the sum of unquenched sensitive-gas deposits. Reference unity is an energy-budget line, not a TRIM raw-energy curve. All scored events have deposits. Original option4 has 26/5,000 events above E0*(1+0.0001): H 0, He 6, C 8, O 7, F 5; screening has zero. The custom QF correction does not rescale raw deposits or native Geant4 electronic response.

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Response fluctuations and explicit recoil cascades

Recorded-response fluctuations and cascade multiplicity

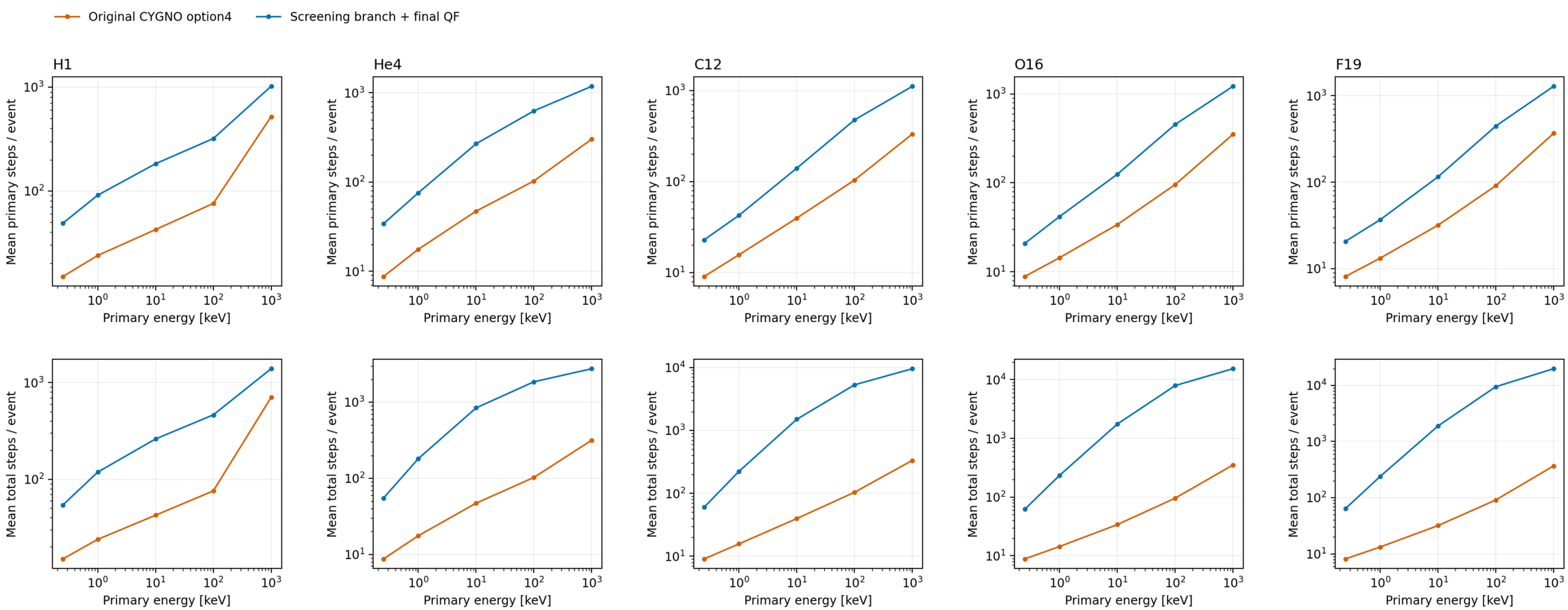


Response CV is the event-level coefficient of variation after each model's custom response treatment. The bounded mapping changes it. There is no retained event-level TRIM electronic-yield distribution for a fluctuation or Fano comparison; no TRIM fluctuation curve is invented. Track multiplicity illustrates explicit cascades and is not itself a detector observable. Head-tail efficiency and reconstructed angular resolution still require digitization validation.

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Computational work indicators

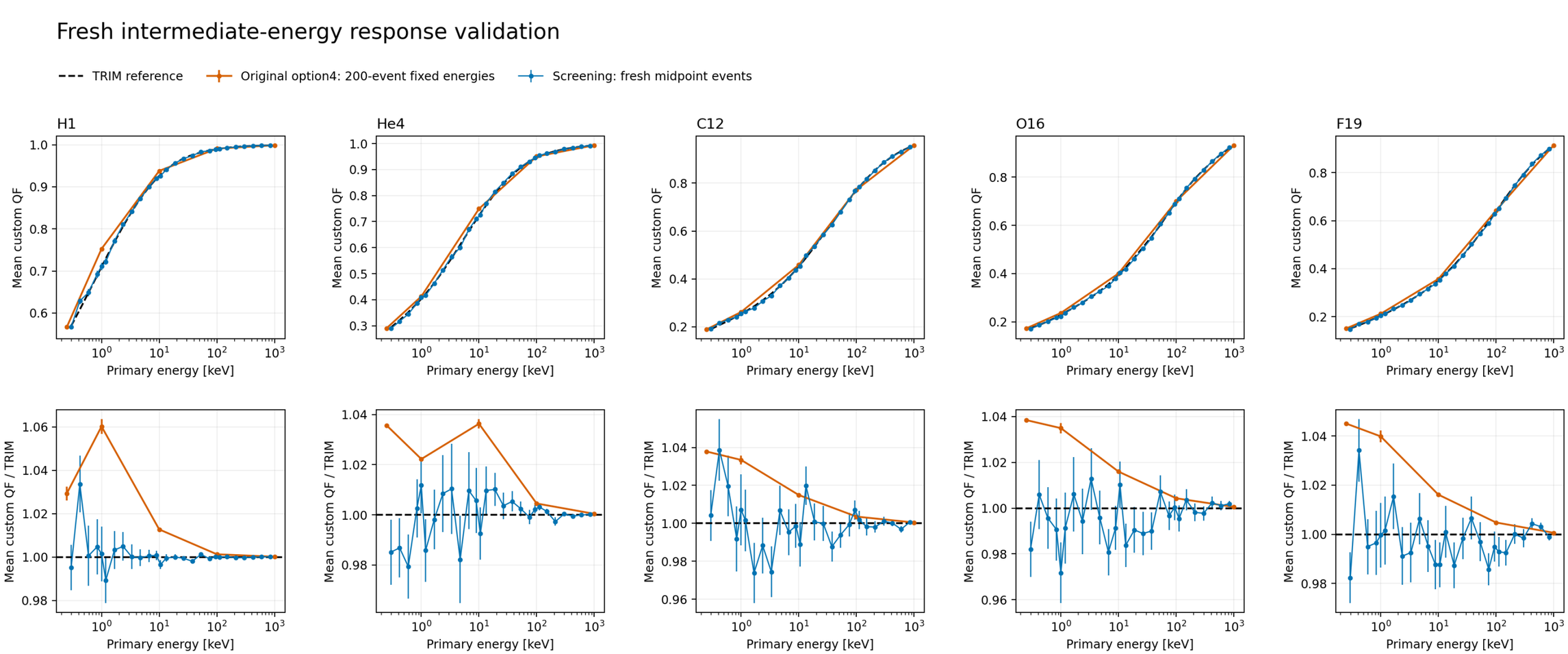
Transport work per event



Screened recoil cascades increase step/track counts, especially for C/O/F. These counts are transport-work indicators, not wall-time speedups or memory benchmarks. A matched production wall-time/peak-memory benchmark is not supplied. ROOT storage is assessed separately on the storage page.

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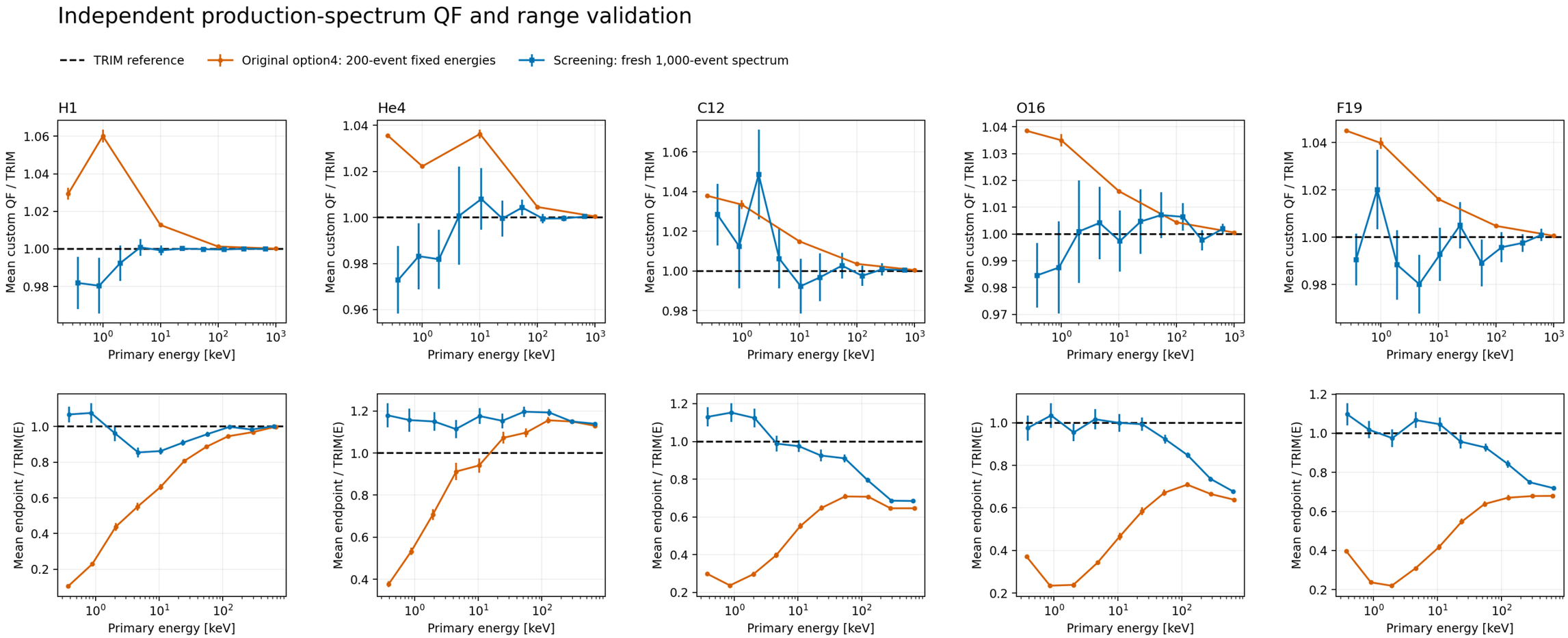
Fresh intermediate-energy response validation



Independent validation after the final dense fit: 27 intermediate energies x 150 events x 5 ions = 20,250 events. Median absolute relative QF error is H 0.056%, He 0.537%, C 0.467%, O 0.608%, F 0.502%. Worst point: 3.872%. Errors include event and conditional calibration statistics. These validation events were not used to fit alpha; model/interpolation systematics are excluded.

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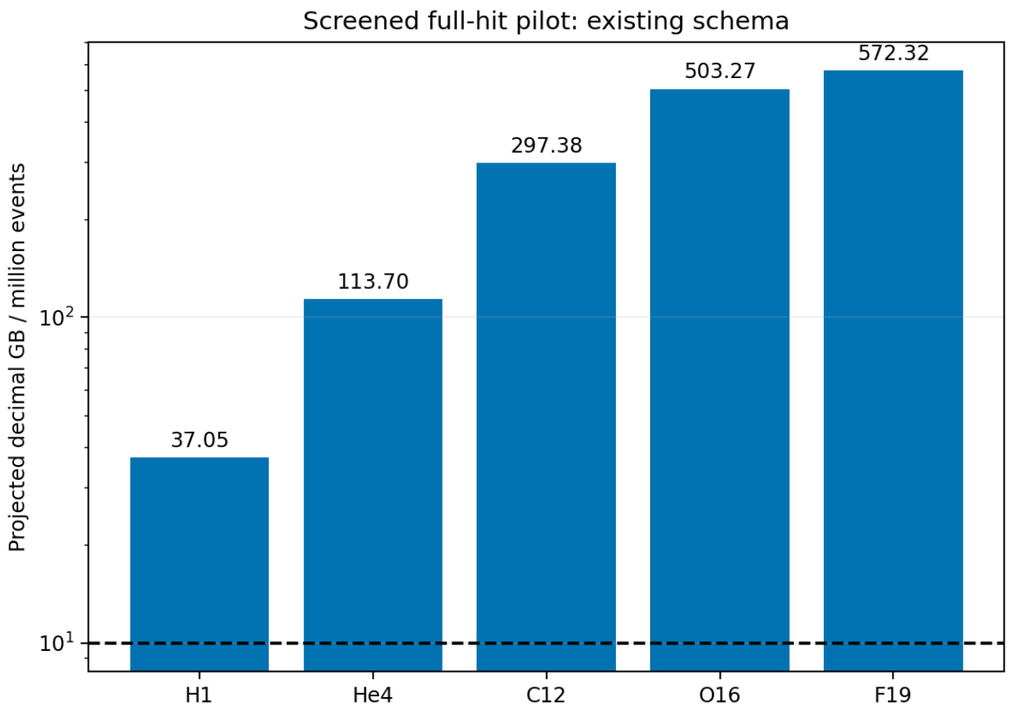
Independent production-spectrum QF and range validation

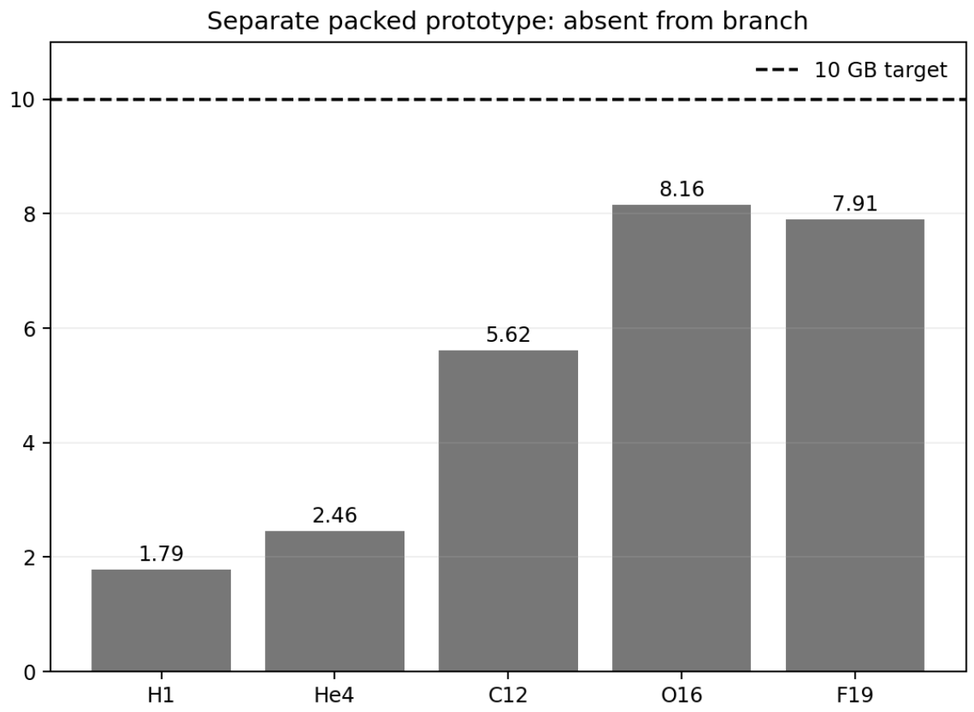


Bottom row: original and screening transport-matched 1,000-event spectra; orange and blue. Bars: MC SEM only.

The final-QF spectrum uses 1,000 fresh log-uniform events/ion. Maximum absolute QF-bin discrepancies are H 1.96%, He 2.71%, C 4.87%, O 1.54%, F 2.02%. QF errors include shared-node conditional calibration uncertainty; the original QF controls remain fixed-energy points. The range panel uses earlier transport-matched full-hit pilots, not another independent run of the integrated branch. Range reference/model uncertainty is excluded in that panel.

ROOT storage: the minimal physics branch does not satisfy 10 GB





Both panels extrapolate 1,000-event/ion pilots to one million events; neither measures a million-event file. The native pilot used identical final transport with the earlier response weights: its byte sizes are approximate for this branch. Packed final-response files preserve event totals/truth but omit raw/native per-hit arrays and aggregate/quantize charge geometry. Their 1.79-8.16 GB projections do not describe this branch's unchanged full ROOT schema. Packed shape guards test storage fidelity, not TRIM head-tail or charge-profile physics. No matched original full-schema storage pilot is supplied here.